

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering and Information Systems (SCORE)

Fall Semester 2026-2027

Assessment - 08

Programme/Branch: M.Tech Software Engineering

Faculty Name: RAMALINGAM M

Course Code: ISWE406P

Course Name: Agile Development Process and

Devops Lab

Slot: L45+L46

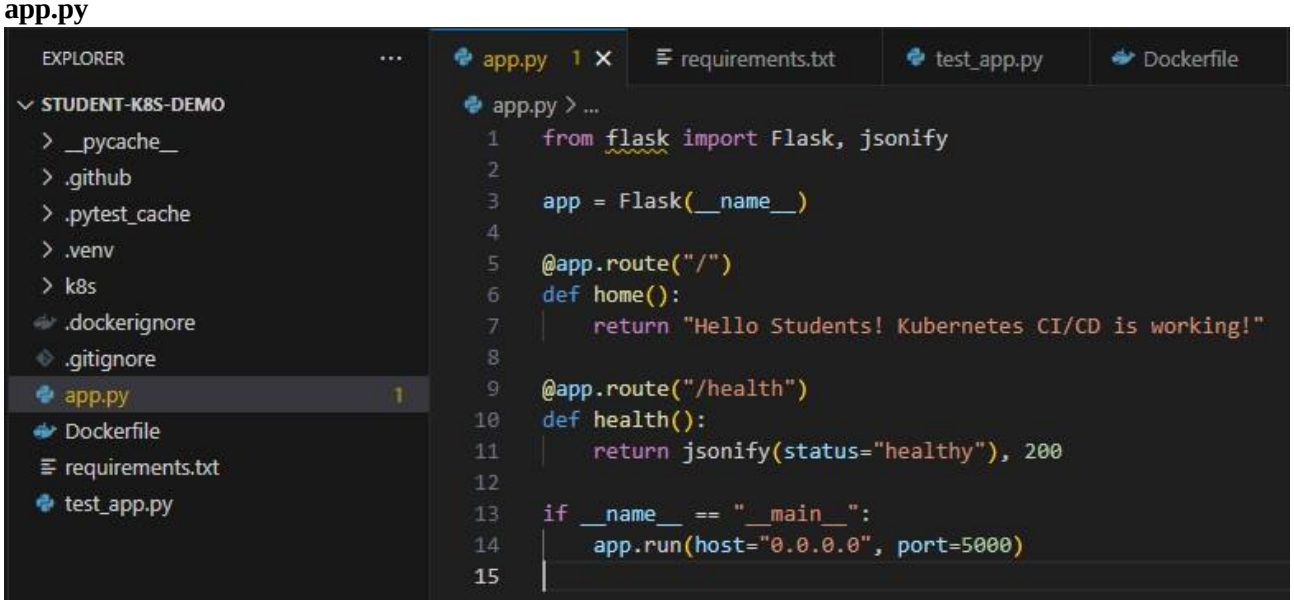
Name: Renu kumar M

Reg No: 24MIS0227

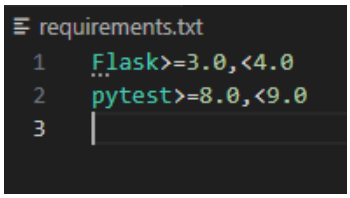
Kubernetes Integration In A CI/CD Pipeline

CODE:

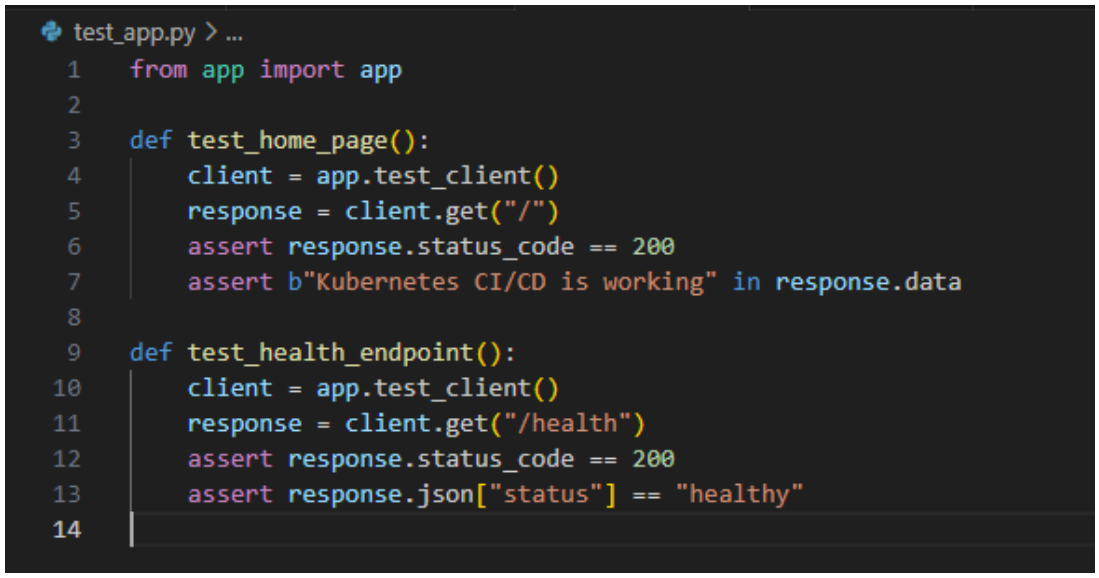
app.py



requirements.txt



test_app.py



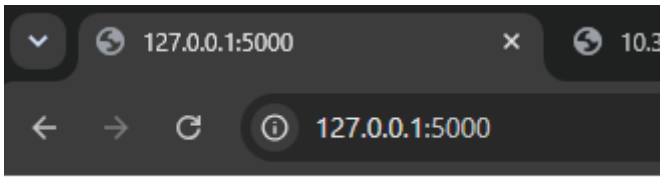
Running Locally:

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> python -m venv .venv
>> .\.venv\Scripts\python.exe -m pip install -r requirements.txt
>> .\.venv\Scripts\python.exe -m pytest -v
>> .\.venv\Scripts\python.exe app.py
>>
Collecting Flask<4.0,>=3.0
  Downloading flask-3.1.3-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 KB 2.0 MB/s eta 0:00:00
Collecting pytest<9.0,>=8.0
  Downloading pytest-8.4.2-py3-none-any.whl (365 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 365.8/365.8 KB 5.6 MB/s eta 0:00:00
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 KB 8.3 MB/s eta 0:00:00
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.9-py3-none-any.whl (228 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 228.7/228.7 KB 6.8 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
  Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Collecting pygments>=2.7.2
  Downloading pygments-2.21.0-py3-none-any.whl (1.3 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.3/1.3 MB 7.9 MB/s eta 0:00:00
Collecting iniconfig>=1
  Downloading iniconfig-2.3.1-py3-none-any.whl (7.6 kB)
Collecting pluggy<2,>=1.5
  Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
up-1.3.1 iniconfig-2.3.1 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.4 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pytest-8.4.2 tomli-2.5.0 typing-extensions-4.16.0 werkzeug-3.1.9
WARNING: You are using pip version 22.0.4; however, version 26.2.1 is available.
You should consider upgrading via the 'C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo\.venv\Scripts\python.exe -m pip install --upgrade pip' command.
```

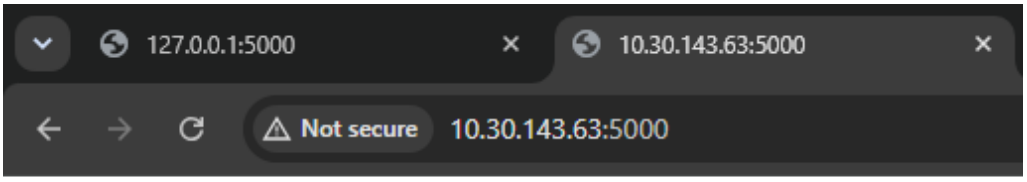
```
up-1.3.1 iniconfig-2.3.1 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.4 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pytest-8.4.2 tomli-2.5.0 typing-extensions-4.16.0 werkzeug-3.1.9
WARNING: You are using pip version 22.0.4; however, version 26.2.1 is available.
You should consider upgrading via the 'C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo\.venv\Scripts\python.exe -m pip install --upgrade pip' command.
============================= test session starts =============================
platform win32 -- Python 3.10.5, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo
collected 2 items

test_app.py::test_home_page PASSED                                       [ 50%]
test_app.py::test_health_endpoint PASSED                                 [100%]

============================== 2 passed in 0.32s ==============================
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.30.143.63:5000
Press CTRL+C to quit
127.0.0.1 - - [14/Oct/2026 16:41:32] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Oct/2026 16:41:32] "GET /favicon.ico HTTP/1.1" 404 -
10.30.143.63 - - [14/Oct/2026 16:41:48] "GET / HTTP/1.1" 200 -
10.30.143.63 - - [14/Oct/2026 16:41:48] "GET /favicon.ico HTTP/1.1" 404 -
```



Hello Students! Kubernetes CI/CD is working!



Hello Students! Kubernetes CI/CD is working!

Dockerfile

```
Dockerfile
1   FROM python:3.12-slim
2   WORKDIR /app
3   COPY requirements.txt .
4   RUN pip install --no-cache-dir -r requirements.txt
5   COPY app.py .
6   EXPOSE 5000
7   CMD ["python", "app.py"]
8
```

.dockerignore

```
.dockerignore
1   .venv/
2   __pycache__/
3   .pytest_cache/
4   .git/
5
```

Run Locally

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> docker build -t student-k8s-demo:v1
>> docker images
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
                                                
```

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> docker stop student-demo
student-demo
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo>
```

Pushing the code Into Github:

.gitignore

```
.gitignore
1   .venv/
2   __pycache__/
3   .pytest_cache/
4   .env
5
```

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git init
Reinitialized existing Git repository in C:/Users/batch1.VITUNIVERSITY/Documents/student-k8s-demo/.git/
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git add .
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git commit -m "Docker"
[master (root-commit) 07d096e] Docker
 9 files changed, 44 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .github/workflows/pipeline.yml
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 k8s/deployment.yaml
 create mode 100644 k8s/service.yaml
 create mode 100644 requirements.txt
 create mode 100644 test_app.py
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git branch -M main
```

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git remote add origin https://github.com/Renu-Kumar-M/docker.git
>> git push -u origin main
error: remote origin already exists.
info: please complete authentication in your browser...
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 20 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (12/12), 1.24 KiB | 634.00 KiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Renu-Kumar-M/docker.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo>
```

pipeline.yml

.github > workflows > ! pipeline.yml

```yaml
name: Student Kubernetes CI/CD
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

permissions:
  contents: read

jobs:
  test:
    name: Checkout and Test
    runs-on: ubuntu-latest
    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt

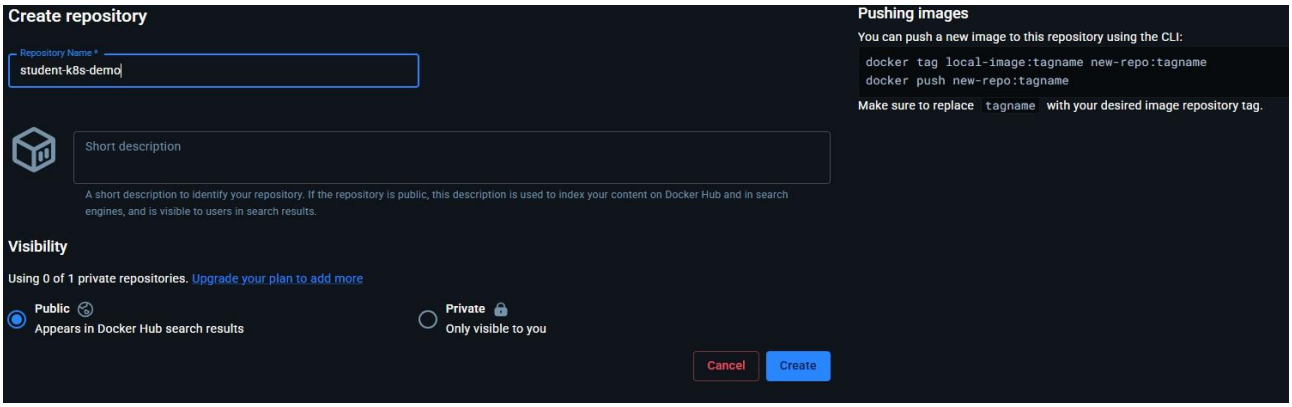
      - name: Run automated tests
        run: python -m pytest -v

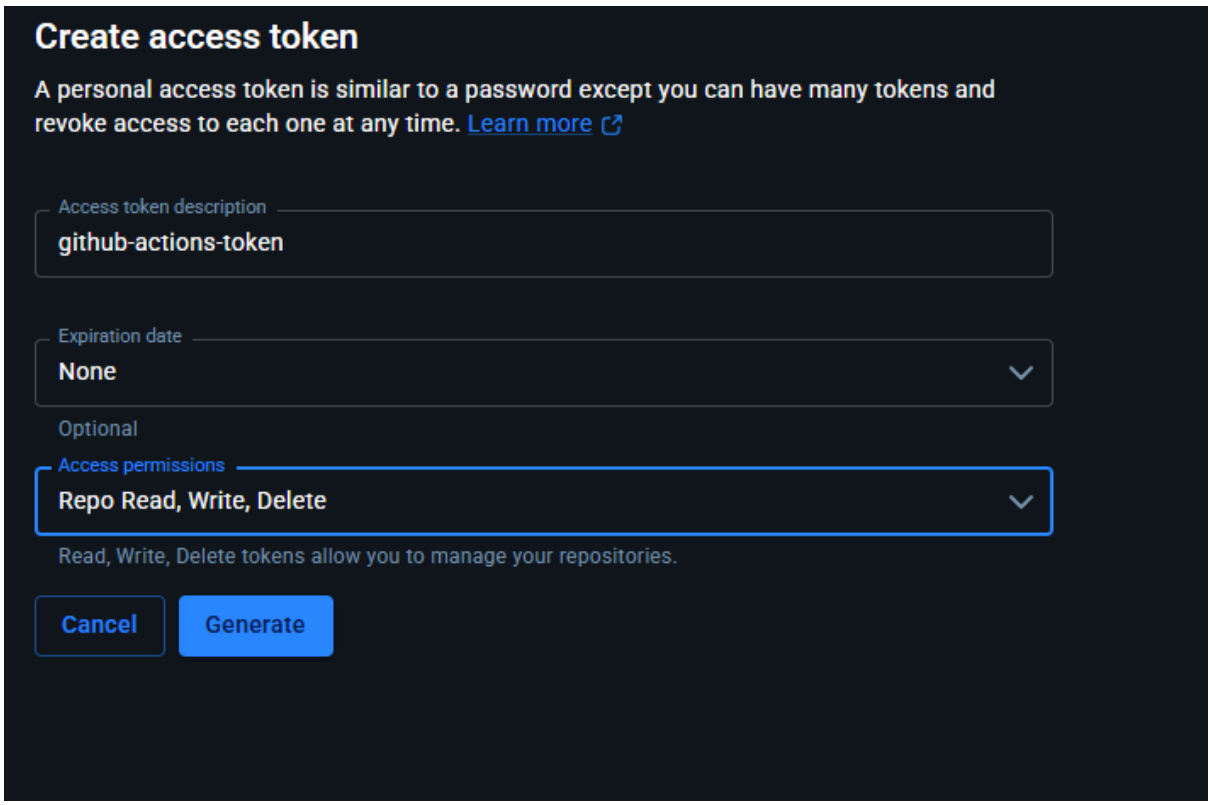
```

Pushing Changes To Github

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo> git add .
>> git commit -m "Add automated CI tests"
>> git push
>>
[main 4f831d2] Add automated CI tests
 1 file changed, 30 insertions(+)
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 663 bytes | 663.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Renu-Kumar-M/docker.git
   07d096e..4f831d2  main -> main
PS C:\Users\batch1.VITUNIVERSITY\Documents\student-k8s-demo>
```

Creating A Repo In Docker:





Modified pipeline.yml

.github > workflows > ! pipeline.yml

```yaml
name: Student Kubernetes CI/CD
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

permissions:
  contents: read

jobs:
  test:
    name: Checkout and Test
    runs-on: ubuntu-latest
    steps:
      - name: Checkout source code
        uses: actions/checkout@v4
      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt
      - name: Run tests
        run: python -m pytest -v

  docker:
    name: Build and Push Docker Image
    needs: test
    if: github.event_name == 'push'
    runs-on: ubuntu-latest
    steps:
      - name: Checkout source code
        uses: actions/checkout@v4
      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v3
      - name: Login to Docker Hub
        uses: docker/login-action@v3
        with:
          username: ${{ secrets.DOCKERHUB_USERNAME }}
          password: ${{ secrets.DOCKERHUB_TOKEN }}
      - name: Build and push image
        uses: docker/build-push-action@v6
        with:
          context: .
          push: true
          tags: ${{ secrets.DOCKERHUB_USERNAME }}/student-k8s-demo:${{ github.sha }}
```

deployement.yaml

k8s > ! deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: student-k8s-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: student-k8s-demo
  template:
    metadata:
      labels:
        app: student-k8s-demo
    spec:
      containers:
      - name: student-k8s-demo
        image: renukumarm06/student-k8s-demo:12dd1ba8903567236986079b127e5715a71fb1e3
        ports:
        - containerPort: 5000
        readinessProbe:
          httpGet:
            path: /health
            port: 5000
          initialDelaySeconds: 5
          periodSeconds: 5
```

service.yaml

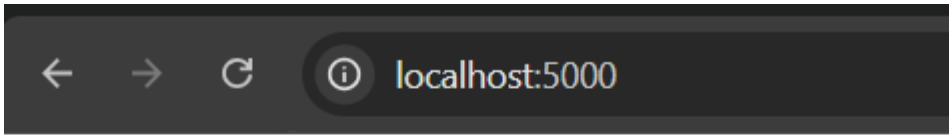
k8s > ! service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: student-k8s-demo
spec:
  type: NodePort
  selector:
    app: student-k8s-demo
  ports:
    - port: 80
      targetPort: 5000
      nodePort: 30080

```

Terminal:

```
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> kubectl apply -f k8s/deployment.yaml
>>
deployment.apps/student-k8s-demo unchanged
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> kubectl apply -f k8s/service.yaml
>>
service/student-k8s-demo unchanged
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> kubectl rollout status deployment/student-k8s-demo
>>
deployment "student-k8s-demo" successfully rolled out
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> kubectl get pods
>>
NAME                               READY   STATUS    RESTARTS   AGE
student-k8s-demo-86897dcbdd-c6mwc   1/1     Running   0          10m
student-k8s-demo-86897dcbdd-vjdkr   1/1     Running   0          10m
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo>
```

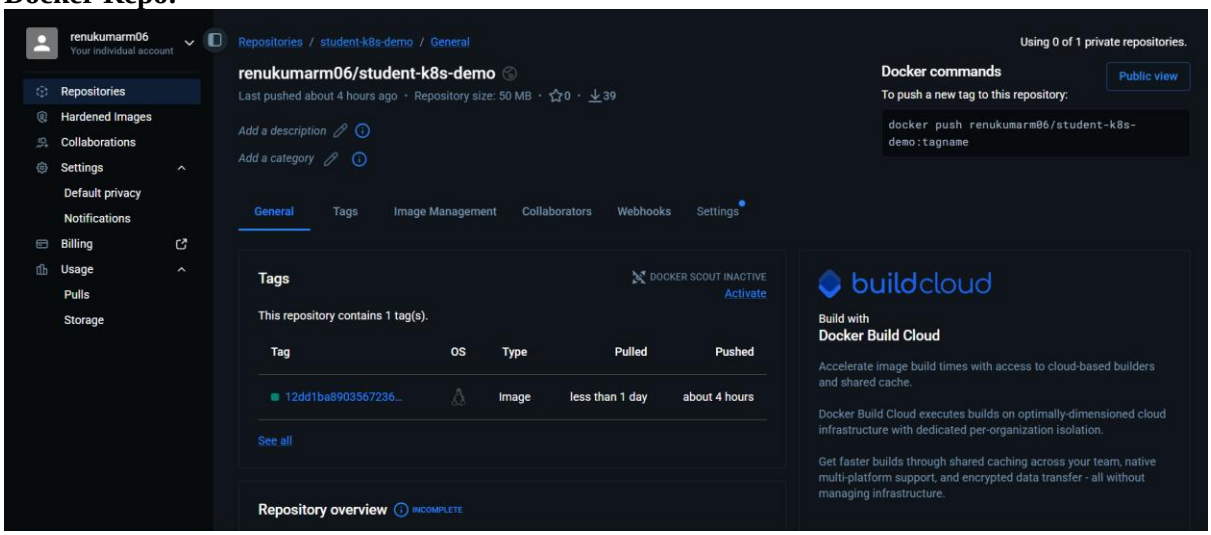


Hello Students! Kubernetes CI/CD is working!

Pushing changes into Github:

```
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> git add .
warning: in the working copy of 'student-k8s-demo/.dockerignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/.github/workflows/pipeline.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/app.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'student-k8s-demo/test_app.py', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> git commit -m "update"
[main c933182] update
 9 files changed, 130 insertions(+)
 create mode 100644 student-k8s-demo/.dockerignore
 create mode 100644 student-k8s-demo/.github/workflows/pipeline.yml
 create mode 100644 student-k8s-demo/.gitignore
 create mode 100644 student-k8s-demo/Dockerfile
 create mode 100644 student-k8s-demo/app.py
 create mode 100644 student-k8s-demo/k8s/deployment.yaml
 create mode 100644 student-k8s-demo/k8s/service.yaml
 create mode 100644 student-k8s-demo/requirements.txt
 create mode 100644 student-k8s-demo/test_app.py
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo> git push -u origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (15/15), 2.30 KiB | 785.00 KiB/s, done.
Total 15 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Renu-Kumar-M/pro.git
   a8672d0..c933182  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Renu Kumar M\OneDrive\Documents\student-k8s-demo>
```

Docker Repo:


renukumarm06
Your individual account
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories / student-k8s-demo / General
Using 0 of 1 private repositories.
renukumarm06/student-k8s-demo
Last pushed about 4 hours ago · Repository size: 50 MB · 0 · 39
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push renukumarm06/student-k8s-demo:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 1 tag(s).
Tag
OS
Type
Pulled
Pushed
12dd1ba8903567236...
Image
less than 1 day
about 4 hours
See all
Repository overview
INCOMPLETE
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.
Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.